

BINYAN XU

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Education

The Chinese University of Hong Kong Ph.D. student in Information Engineering	Sep. 2023 – Aug. 2027 GPA: 3.86
University of California, Berkeley Exchange Program in EECS	Aug. 2021 – Dec. 2021 GPA: 4.0
Xi'an Jiaotong University B.Sc. in Automation	Sep. 2019 – Jul. 2023 Honor, GPA: 3.92

Publications (First Author)

- Binyan Xu et al.**. From Internal Diagnosis to External Auditing: A VLM-Driven Paradigm for Data-Free Online Backdoor Defense. In *ICML '26*, Under Review (avg. score 4.25).
- Binyan Xu et al.**. From Multi-Agent to Single-Agent: When Is Skill Distillation Beneficial? In *NeurIPS '26*, Under Review.
- Binyan Xu et al.**. Breaking the Stealth-Potency Trade-off in Clean-Image Backdoors with Generative Trigger Optimization. In *AAAI '26*, 2026, **Oral Presentation (CCF-A)**.
- Binyan Xu et al.**. One Surrogate to Fool Them All: Universal, Transferable, and Targeted Adversarial Attacks with CLIP. In *CCS '25*, 2025, **Oral Presentation (CCF-A)**.
- Binyan Xu et al.**. CLIP-Guided Backdoor Defense through Entropy-Based Poisoned Dataset Separation. In *MM '25*, 2025, **Oral Presentation (CCF-A)**.

Internships

- Agentic AI Research on Single-Agent and Multi-Agent Systems** **Jan. 2026 – Present**
Tencent, Qingyun Program Internship | LLM Agent, Skill Distillation, Multi-Agent System
- Studied when distilling a multi-agent system (MAS) into a single-agent skill yields performance gains.
 - Proposed Metric Freedom, a principled predictor of skill utility measured before any skill is built.
 - Adaptive skill matches or exceeds original MAS at 1.4–15× lower cost across 4 tasks and 11 datasets.
 - 1 paper submitted to NeurIPS (Under Review).
- AI-Aided Ascetic Graphic Design Generation** **Jul. 2022 – Jun. 2023**
Microsoft Research Asia (MSRA), Stars of Tomorrow | Diffusion Model, Large Language Model
- Developed an ascetic graphic design system with an autoregressive vision-language model for layout generation.
 - Enhanced this graphic design system with diffusion model to generate additional decorative patterns.
 - Significantly outperforms previous approaches in both better generation quality and less constraint violation.

Projects

- Foundation Model-Aided Neural Network Attacks and Defenses** **Sep. 2023 – Present**
Laboratory for Applied Security Research, CUHK | Backdoor, Adversarial Attack, VLM, Diffusion Model
- Proposed an InfoGAN-based stealthy backdoor attack, achieving $\geq 90\%$ ASR with $\leq 1\%$ clean accuracy drop.
 - Involved in UnivIntruder, a transferable attack achieving 99.4% ASR across 4 datasets and 84% on Google/Baidu.
 - Proposed CLIP-guided backdoor defense via entropy-based data separation, defending all kinds of backdoor attacks.
 - Proposed a VLM-driven data-free online backdoor defense paradigm leveraging VLMs as external auditors.
 - 3 papers accepted (CCS '25, MM '25, AAAI '26, all CCF-A Oral); 1 paper submitted to ICML '26 (avg. score 4.25).

Selected Awards

- Finalist Award, Mathematical Contest in Modeling
- Outstanding Graduates Scholarship
- Qian Xuesen Program Honorary Graduate
- First-class Funding for Studying Abroad (about ¥100k)

Skills

Languages: Chinese (Native), English (Fluent); TOEFL: 104 (R29/L28/S22/W25), CET-6: 584
Programming: Python, PyTorch, TensorFlow, C/C++, MATLAB